Green Building Science Test

Name: _____ Class Section: _____

[WINDOW_DIRECTION]

1. You are designing a house for a client in Boston, Massachusetts. The client would like to have a large window on one side of the house. There are no trees or other buildings around this house. To maximize the energy efficiency of the house in the winter, on which side of the house would you choose to install the large window?

a) East side
b) West side
c) North side
d) South side
e) I am not sure

Please explain your choice:

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Answer key: D. The large south-facing window will be most energy efficient in the winter. In Chicago, which is in the northern hemisphere, the sun rises in the southeast, peaks out at a low angle above the southern horizon, and then sets in the southwest. Because the sun is on the south side of the house all day long, the south-facing window will receive most heat through thermal radiation. The larger the window, the more heat gain. Thus, having the large window on the south wall is the most energy efficient choice.

[NOON_SHADE]
2. A house is built on a flat lot with no trees or other buildings around in Boston, Massachusetts. Which side of the house would be shaded most at NOON regardless of the time of the year?

a) East side
b) West side
c) North side
d) South side
e) All sides are equally shaded
f) I am not sure

Please explain your choice:

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Answer key: C. The north side of the house will be shaded most at noon regardless of time of the year. In Boston, which is in the northern hemisphere, the sun rises in the southeast, peaks out at a low angle above the southern horizon, and then sets in the southwest. Thus at noon, the house blocks sunlight from shining on the north wall.

[ROOF_COLOR]
3. Which of the following colors would you choose for the WALLS of your house in order to reduce your cooling cost in the summer?

a) Black
b) Dark gray
c) Light gray
d) Color doesn't matter
e) I am not sure

Please explain your choice:
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Answer key: C. Light colored roof would be more energy efficient. Light color reflects more light than dark color, thus it absorbs less heat and stays relatively cooler. The cooler roof in the summer reduces the temperature difference between the roof and the inside, which in turns slow down heat transfer from the inside to the roof. As a result, the energy is required to cool the house.

[WINDOW_U_VALUE]

4. If you were to replace a large single-pane window (U-factor: 5.91 W/(m^2. °C)) in your house with a double-pane one of the same size (U-factor: 3.12 W/(m^2. °C)), what will happen to the energy required to cool your house in the summer?

a) It will stay the same
b) It will increase
c) It will decrease
d) I am not sure

Please explain your choice:

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Answer key: C. The energy required to cool the home will decrease. The U-factor is a measure of the rate of heat loss of a particular building material. The smaller the U-factor, the slower heat transfers through the window, requiring less energy to keep the house cool.

[SOLAR_SIDE]
5. For a house located in Boston, Massachusetts, which side of the roof would you choose to install solar panels in order to maximize their ability to convert solar energy to electricity?

a) East side
b) West side
c) North side
d) South side
e) I am not sure

Please explain your choice:
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Answer key: D. The sunlight in the northern hemisphere is mostly from the south. Solar panels need to receive direct sunlight to maximize their efficiency. Thus, installing solar panels on the south side of the roof will maximize their efficiency.

[SOLAR_ANGLE]
6. For a house located in Miami, Florida, on which type of roof would solar panels be most efficient?

a) A steep roof
b) A slightly sloped roof
c) A flat roof
d) The angle of the roof does not matter
e) I am not sure

Please explain your choice:
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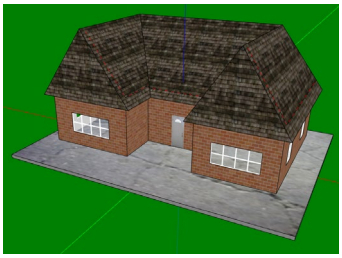
Answer key: B. A slightly sloped roof would maximize the efficiency of solar panels. The sun is higher in locations closer to the equator. A slightly sloped roof would increase the amount of insolation, which in turn improves the efficiency of installed solar panels.

[HOUSE_SHAPE]

7. The three houses below have identical square footage, windows, and roof height and material. The only difference among them is their shape. In a WINTER NIGHT, which house would lose the most heat?

Rectangle

L-shaped



U-shaped

a) The rectangle house
b) The L-shaped house
c) The U-shaped house
d) They would lose the same amount of heat
e) I am not sure

Please explain your choice:

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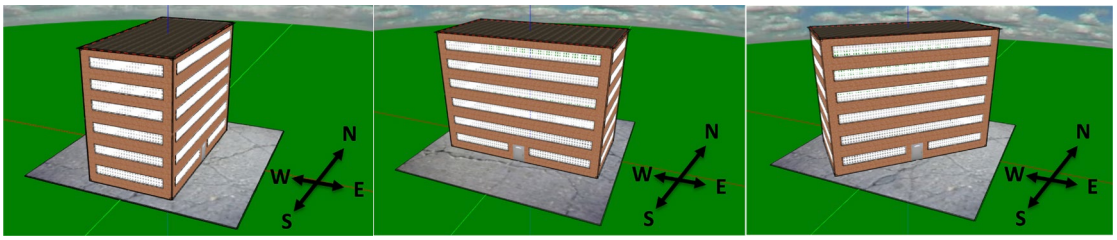
Answer key: C. The U-shaped house loses heat the fastest, the L-shaped house the moderate, and the rectangle house the slowest. According to Fourier's law of thermal conduction, thermal conduction, the large surface area the more heat transfer. With the same square footage, the U-shaped house has the largest surface area, while the rectangle house has the smallest surface area. With all other conditions the same, the U-shaped house loses heat the fastest and the rectangle house loses heat the slowest.

[RECTANGLE_ORIENTATION]

8. A building will be built on a 50m by 50m block with no trees or other buildings around in Boston, Massachusetts. The dimensions of the building are set to be 40m long, 20m wide, and 30m high. Below are three ways to orient the building on the block. Which orientation would allow the building to gain the MOST energy from the sun in the WINTER?



Orientation 1 Orientation 2 Orientation 3

a) Orientation 1
b) Orientation 2
c) Orientation 3
d) The building would absorb the same amount of solar energy no matter how it is oriented.
e) I am not sure.

Please explain your choice:

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Answer key: B. The long axis of the building runs east-west. In the northern hemisphere, where Washington, DC is located, the east and west walls of a building gain most solar energy among all four walls. Setting the long axis of the building along the east-west direction will reduce the surface areas of the east and west walls, resulting in decreases in solar energy gains on these two walls.

[TREE_TYPE]

9. A two-story south-facing house is about 30 feet high and located in Massachusetts. The house owner wants to plant trees to improve energy efficiency of the house. Which of the following trees would you choose, assuming the total costs of these options are the same?

a) Ten boxwood trees, 5 feet tall and 4 feet wide, Evergreen (doesn't shed leaves annually)
b) Ten Japanese barberry trees, 5 feet tall and 4 feet wide, Deciduous (sheds leaves annually)
c) Two Weeping Alaskan Cedar trees, 25 feet tall and 20 feet wide, Evergreen (doesn't shed leaves annually)
d) Two White Dogwood trees, 25 feet tall and 20 feet wide, Deciduous (sheds leaves annually)
e) I am not sure

Please explain your choice:

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Answer key: D. The taller the trees, the more hours of shade they can provide to reduce solar heat gain in the summer. However, in the winter, solar gain needs to be maximized. Thus, deciduous trees are better than evergreen trees because their bare branches block less sunlight during winter.

[TREE_SIDE]
10. For the house described in the last question (south-facing, two-story, located in Massachusetts), on which side of the house would you plant the trees?

a) East
b) Southeast
c) South
d) Southwest
e) West
f) I am not sure

Please explain your choice:

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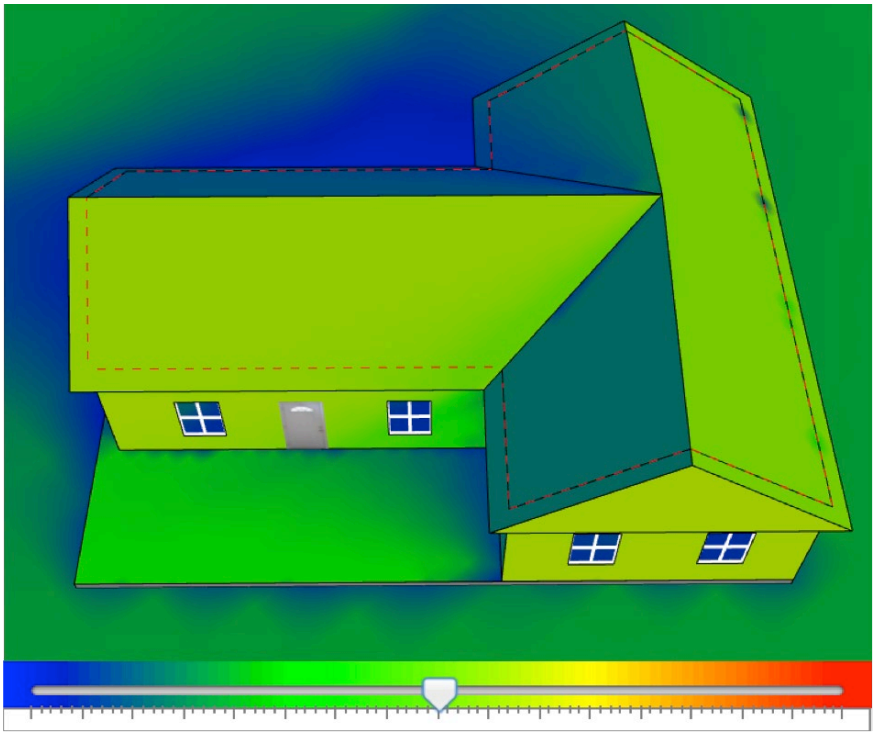
Answer key: E. West side. The west side and east side will receive a lot of heat during summer, and the west side feels even hotter than the east because the afternoon sunshine adds to the existing high temperature at the time. Planting trees on the west side of the house would block much sunlight coming from the west.

[SOLAR_HEAT_MAP]

11. The image below shows the temperature distribution of a house located in Boston, Massachusetts at noontime on a sunny day in January. As shown in the temperature gradient bar below the image, warmer colors (i.e., red side) represent higher temperature, and colder colors (i.e., blue side) represent lower temperature. Which direction does the front side face (the front side has the main door)?



a) East
b) Southeast
c) South
d) Southwest
e) West
f) I am not sure

Please explain your choice:

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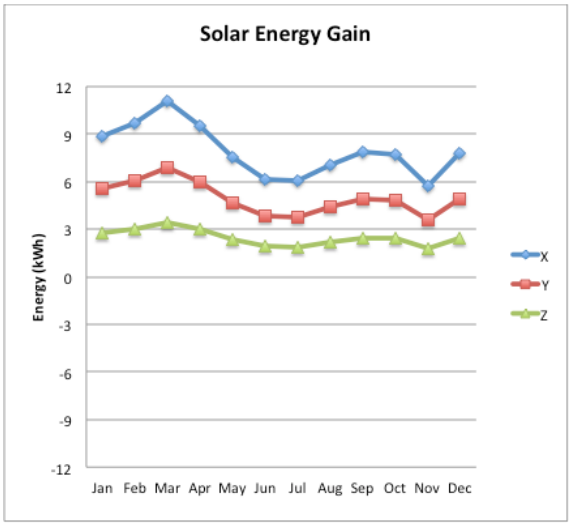
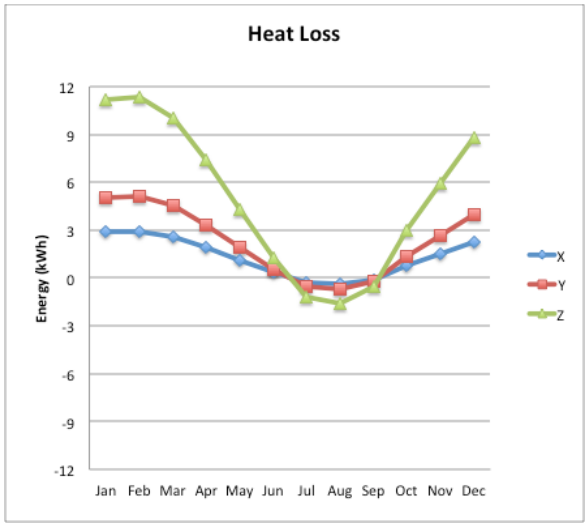
Answer key: D. The house faces southwest. According to the temperature color coding, green is warmer than blue. First, the roof on the front has warmer than that on the back, so the front must face south in general. Second, the right side of the secondary roof is warmer than the left side, so the right side must face the south. Thus, the house front must face southwest to generate both effects.

[WINDOW_GRAPH_S] [WINDOW_GRAPH_U]

12. Three windows of the same size but different insulation and solar properties are installed on the south side of a house located in Boston, Massachusetts. The graphs below show the solar energy gain and heat loss through these three windows throughout the year. In the table below the graphs, please mark the appropriate Solar Heat Gain Coefficient (SHGC) and U-Factor for each window.





	Solar Heat Gain Coefficient (SHGC)			U-Factor W/(m². °C)		
Window X (blue)	☐25%	☐50%	☐80%	☐1.53	☐2.66	☐5.91
Window Y (red)	☐25%	☐50%	☐80%	☐1.53	☐2.66	☐5.91
Window Z (green)	☐25%	☐50%	☐80%	☐1.53	☐2.66	☐5.91
	☐I am not sure			☐I am not sure		

Please explain your choices

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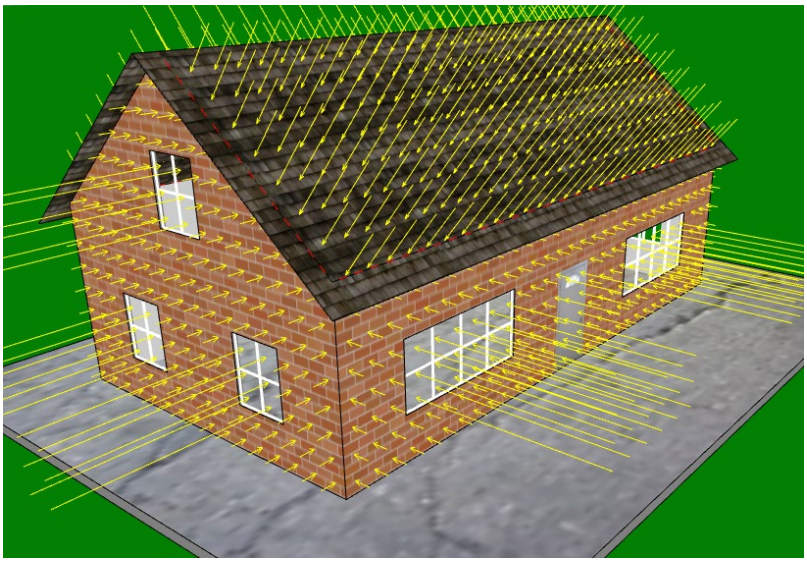
Answer key:
Window X: SHGC 80%, U-Factor 1.53
Window Y: SHGC 50%, U-Factor 2.66
Window Z: SHGC 25%, U-Factor 5.91
Higher SHGC results in higher solar energy gain, thus Window X has the highest SHGC and Window Z has the lowest SHGC. Higher U-Factor means faster heat loss, or greater negative heat gain, thus Window X has the lowest U-factor and Window Z has the highest U-Factor.

[HEAT_FLUX_SEASON]

13. The image below shows the heat flux analysis of a house. The arrows point to the direction of heat transfer, and the length of the arrows represents the rate of heat transfer (longer arrows indicate faster heat transfer). Which season is most likely selected for this heat flux analysis?

a) Spring
b) Summer
c) Fall
d) Winter
e) I am not sure



Please explain your choice:

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Answer key: B. Heat transfers from the outside to the inside of the house, so the outside must be hotter than the inside. Thus, it is most likely to in the summer.

[HEAT_FLUX_U]
14. Based on the heat flux analysis of the house shown in the last question, which building element has the LOWEST U-factor?

a) Walls
b) Windows
c) Doors
d) Roof
e) I am not sure

Please explain your choice:
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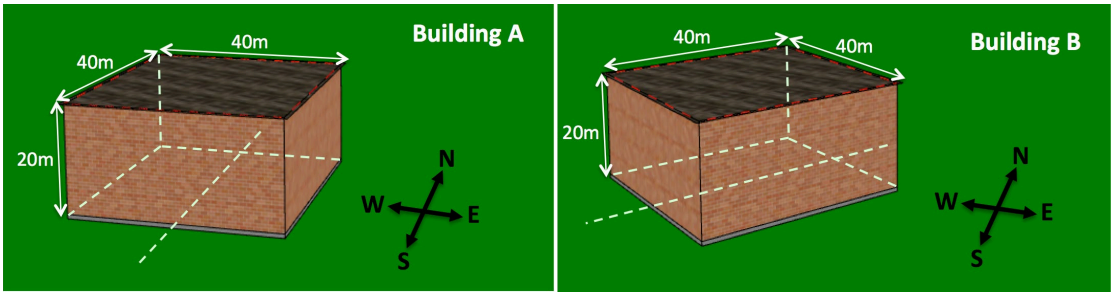
Answer key: A. As the arrows along the walls are the shortest, heat transfer is slowest there. So the walls must have a low rate of heat loss, which is indicated by the U-factor.

[SQUARE_ORIENTATION]

15. The two square buildings below are 40 meters wide, 40 meters long, 20 meters tall, and located in Boston, Massachusetts. They are identical except for their orientations. The axis of building A runs north-south, while the axis of building B runs southeast-northwest. Assuming there are no trees or other buildings around the two buildings, how do their TOTAL solar energy gains THROUGHOUT THE DAY compare?



Building A Building B

a) Building A gains more solar energy than building B.
b) Building A gains less solar energy than building B.
c) They have the same solar energy gain.
d) I am not sure.

Please explain your choice:

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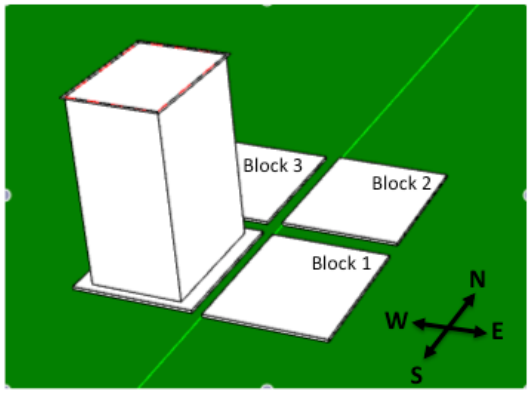
Answer Key 1: B. Larger surface area absorbs more solar energy. While all sides of building B receive sunlight, only three sides of building A do so. Thus, building A gains less solar energy than building B.

Answer Key 2: B. Insolation is determined by the width of the sunlight beam striking on a surface regardless of the angle between the beam and the surface. For both buildings, the width of the sunlight beam fluctuates between the length of the side (minimum) and the length of the diagonal (maximum) over the day. For building A, the width of the sunlight beam reaches the maximum for twice and the minimum for three times. For building B, the width of the sunlight beam reaches the maximum for three times and the minimum for twice. Therefore, the average width of sunlight beam striking on building A is less than that on building B, and it follows that the building A gains less solar energy than building B.

[BLOCK_CHOICE]

16. The four city blocks below are located in Boston, Massachusetts. All of them are 40 meters long and 40 meters wide. There is an existing high-rise building on one of them. You will build another high-rise building similar to the existing one and leave the remaining blocks as open space. You want this new building to have the optimal solar energy gains—as high as possible in the winter and as low as possible in the summer. Which block would you choose for the new building?



a) Block 1
b) Block 2
c) Block 3
d) Any one of them
e) I am not sure

Please explain your choice:

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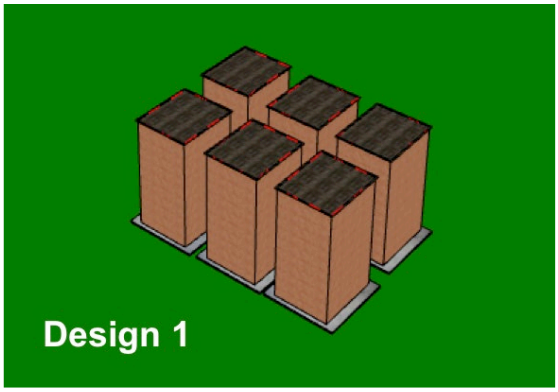
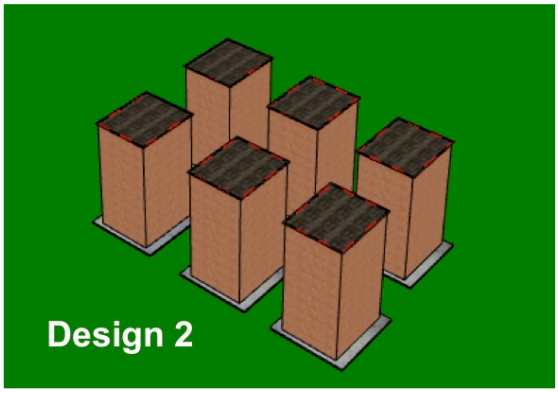
Answer key: A. building built on block 1 will have the optimal solar energy gains. The building on block 1 receives most sunlight on the south wall, which maximizes its solar energy gain in the winter. Meanwhile, the building on block 1 has the west wall completely shaded by the existing building, which minimizes its solar energy gain in the summer.

[URBAN_DENSITY_TOTAL]

17. Below are two urban designs made for a block in the New York City. The six residential buildings in the two designs are identical except that they are closer to each other in Design 1 than in Design 2. How do the total solar energy gains of the two designs compare?





a) The total solar energy gains of the two designs are the same.
b) The total solar energy gain of Design 1 is greater than that of Design 2.
c) The total solar energy gain of Design 1 is less than that of Design 2.
d) I am not sure.

Please explain your choice:

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Answer key: C. The total solar energy gain of Design 1 is less than that of Design 2. With greater distance among buildings, more building surface area can receive sunlight, resulting in an increase in the total solar energy gain.

[URBAN_DENSITY_UNEVEN]

18. For the two urban designs described in the last question, which of the following statements correctly describes how solar energy gains are distributed among the six buildings in the two designs?

a) Solar energy gains are evenly distributed among the buildings within each design.
b) Solar energy gains are unevenly distributed among the buildings within each design, but the degrees of unevenness in the two designs are the same.
c) Solar energy gains are unevenly distributed among the buildings within each design, and the degree of unevenness in Design 1 is greater than that in Design 2.
d) Solar energy gains are unevenly distributed among the buildings within each design, and the degree of unevenness in Design 1 is less than that in Design 2.
e) I am not sure.

Please explain your choice:

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Answer key: C. Solar energy gains are unevenly distributed among the buildings within each design, and the degree of unevenness in Design 1 is greater than that in Design 2. In a cluster of buildings, each building creates shades for others. Depending on their positions within the block, each building has different amount of surface area being shaded. Thus, the solar energy gains are unevenly distributed among them. Greater distance among the buildings reduces the difference in surface area.